MINI PROJECT REPORT

ON

GAN-Based Synthetic Image Generation for Leaf Disease Classification Using Different CNN Models

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In Partial Fulfilment of the Requirement for the Award of

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COMPUTER SCIENCE & ENGINEERING

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CERTIFICATE

This is to certify that the project entitled

"GAN-Based Synthetic Image Generation for Leaf Disease Classification Using Different CNN Models"

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Computer Science and Engineering) at KIIT Deemed to be University, Bhubaneswar. This work is done during the year 2025-2026, under our guidance.

Date: ___/___/_____

(Prof. SUCHISMITA DAS)
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Abstract

Leaf diseases in plants lead to losses of 2040% annual crop-yields in the world. This mini project is a comparative analysis of GAN-based synthetic image generation in the classification of plant leaf disease with four CNN architectures, namely, ResNet50, DenseNet121, EfficientNetB0, and MobileNetV2. All architectures have a standard four-stage pipeline: (1) baseline training on real images with stratified 5-fold cross-validation; (2) standard data augmentation; (3) GAN-based synthetic oversampling (DCGAN or WGAN-GP) to minority classes; and (4) CBAM attention integration on the GAN-balanced model. All models are fine-tuned on 6,218 and 20,638 images of PlantVillage subsets with ImageNet-pretrained weights.

The main findings are: DCGAN augmentation improved MobileNetV2 accuracy by 1.49 pp to 99.55% with Tomato Early Blight recall improving by 22.5% (0.80→0.98); CBAM on denseNet121 improved Apple Scab recall by 4.03 pp; ResNet50 achieved a 100.0000 recall on the The validity of synthetic image quality was tested by FID and KID values with a minimum of 2.65 on Potato Late Blight. Findings prove that GAN-based augmentation is a conceptual solution to solving the problem of class imbalance in detection of agricultural diseases.

Keywords: Plant Disease, GAN, DCGAN, WGAN-GP, ResNet50, DenseNet121, EfficientNetB0, MobileNetV2, CBAM, Transfer Learning, Class Imbalance, PlantVillage, FID, KID

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Introduction

Background and Motivation

Food security of the whole human population and over two billion livelihoods all over the world are supported by agriculture. Plant diseases caused by fungal pathogens, bacteria, viruses, and environmental stress contribute to an estimated 2040% of the total loss in crop-yields globally every year. Agronomic intervention must be timely through early and accurate detection of the disease symptoms. Conventional methods of detection are based on the practice of trained agronomists conducting inspections in the fields which is slow, costly and inapplicable to the contemporary farming activities. Smallholder farmers can hardly access expertise, which results in delayed treatment, excessive use of pesticides, and avoidable economic losses.

Convolutional Neural Networks (CNNs) have revolutionized automated image classification with deep learning. The pre-trained architectures ResNet50, DenseNet121, EfficientNetB0, and MobileNetV2, trained on ImageNet (1.2 million images), exhibit impressive transfer learning performance. When applied to the PlantVillage benchmark, the models have accuracy of over 95 percent, making it technically feasible to detect plant diseases with automated methods. But the situation is complicated in practice by class imbalance: rare disease groups can only have a few hundred training images, whereas common classes can have thousands, leading to the under-learning of rare diseases by the classifier.

The Class Imbalance Problem

Bias in the presence of classes imbalance is that CNN classifiers are biased towards majority classes, causing an artificial high overall accuracy and collapse in minority class recall. Imbalanced ratios in our datasets have a range of 2.6:1 (DenseNet121 subset) to 21:1 (MobileNetV2 subset), with the most extreme being Potato Healthy with 129 images and Tomato Yellow Leaf Curl Virus with 2,727 images. Traditional augmentation uses geometric and photometric transformations on the existing images - useful in regularizing but incapable of altering the underlying distribution of classes. The GANs create completely new images using the learned data distribution of every minority class and introduce new informational content. This project explores and compares the two methods on four CNN architectures.

Generative Adversarial Networks

GANs were proposed by Goodfellow et al. (2014) as a min-max adversarial game between a Generator G and a Discriminator D, and DCGAN (Radford et al., 2015) stabilised training using convolutional architectures and batch normalization, which can be trained with minority classes containing at least a few thousand real training examples. WGAN (Arjovsky et al., 2017) replaced Binary Cross-Entropy loss with the Wasserstein distance, maintaining informative gradients on small datasets. WGAN-GP (Gulrajani et al., 2017) implements the Lipschitz constraint by using gradient penalty (10) allowing it to be trained relatively smoothly even when using 129 real images (DCGAN fails to learn at all).

CBAM Attention Mechanism

The Convolutional Block Attention Module (Woo et al., 2018) applies sequential channel and spatial attention to CNN feature maps. Channel attention identifies which feature maps matter; spatial attention identifies where in the image disease symptoms appear. Both produce sigmoid-gated weight maps multiplied element-wise with feature maps, directing the network's focus to discriminative leaf regions. CBAM is inserted after the final convolutional block of each backbone, before global average pooling.

Project Objectives

- Establish baseline CNN performance for all four architectures using stratified 5-fold cross-validation.

- Evaluate the impact of conventional augmentation on each model.
- Train class-conditional GAN models on minority classes and test the evaluation of synthetic image quality using FID and KID.
- Retrain each CNN on GAN-augmented balanced datasets and measure per-class recall improvements.
- Integrate CBAM into each backbone and assess minority class recall gains.

Dataset Overview

The PlantVillage data are utilized in all the models as subsets (Hughes & Salathé, 2015). Member 1 and Member 3 are on a 6-class subset (Pepper-Potato-Tomato); Member 2 on a 6-class subset (Apple- Cherry-Corn); Members 4 and 5 on a 15-class multi-species subset. Stratified partitions that use a fixed random seed 42 make partitions replicable. Only real images exist in test sets: synthetic images can be restricted to training sets to avoid data leakage. Normalizing images to ImageNet statistics and resizing them to 224x224 are attained by all models.

Report Organization

The report is structured by the following: Basic Concepts and Literature review, which deals with basic architectures and methods. Problem Statement documents the system requirements and research problem. Details of implementation cover all four model pipelines, experimental details, quantitative outcome, and quality assessment of GAN. Standards Adopted are design, coding, and testing standards. Future Scope and Conclusion summarizes results. All the works that have been cited are listed in the reference list. The Individual Contribution section pays attention to the individual contributions of the members, on the technical side.

Basic Concepts and Literature Review

Convolutional Neural Networks

CNNs learn to obtain spatially local features using learnable convolutional filters, project representations by the pooling mechanism and transform the representations to class probabilities through fully connected networks. The CNNs invented by AlexNet (Krizhevsky et al., 2012) set the new paradigm on ImageNet, reducing the error of the top-5 to 15.3 percent compared to 26.2 percent of the previous state of the art with an increase of accuracy and a reduction in the number of parameters. CNNs were subsequently enhanced with batch normalization, residual connections, dense connectivity, compound scaling ImageNet-pretrained CNNs are now the rod in the ground in terms of transfer learning to domain-specific classification in the setting of limited labeled data..

ResNet50 – Residual Learning

ResNet was introduced by He et al. (2016), which addresses the issue of vanishing gradient by shortcut connections, which produce identity mappings between residual blocks. The ResNet50 includes 50 layers of bottlenecks with residual blocks (1×1 Convolution into 3×3 convolution into 1×1 convolution) making a total of nearly 25.6 million parameters and ImageNet top-1 accuracy of 76.1%. Mohanty et al. (2016) used the entire PlantVillage dataset of 38 classes with ResNet, which had a 99.35% accuracy. Member 1 applies ResNet50 to the 6-class Pepper-Potato-Tomato subset (6,218 images) using WGAN-GP augmentation and CBAM attention in this project..

DenseNet121 – Dense Connectivity

Huang et al. (2017) presented DenseNet with direct input of feature maps of each layer on the previous layer. This high connectivity encourages feature reuse, technology, improved gradient flow and minimizes redundant parameter learning. DenseNet121 is made up of four dense blocks (6, 12, 24, 16 layers) with a total channel output capacity of 1,024. Dense connection works well in classification of leaf diseases as patterns of disease appear at several spatial resolutions at the same time. Member 2 uses DenseNet121 to the Apple-Cherry-Corn 6-class subset (6,535 images) in DCGAN and CBAM.

EfficientNetB0 – Compound Scaling

EfficientNet, proposed by Tan and Le (2019), scales depth, width, and resolution of CNNs jointly with fixed compound coefficient of φ. EfficientNetB0 requires only 5.3 million parameters to give comparable accuracy to ResNet50 - 79 percent fewer parameters. In the form of Mobile-Inverted Bottleneck (MBConv) blocks, it is intrinsically feature-selective, feature-squeeze-excitation attention is used, and the depthwise separable convolutions, as well as the desirable feature selectivity and equilibrium between parameters and features, make it effectively parameter-efficient. Member 3 uses EfficientNetB0 to the 6-class Pepper-Potato-Tomato (9,546 images) dataset using WGAN-GP and CBAM.

MobileNetV2 – Lightweight Architecture

MobileNetV2 is tailored by Sandler et al. (2018) to be deployed to mobiles and embedded systems with just 3.4 million parameters. It uses inverted residual blocks to increase the feature dimensions in depthwise convolution space with linear bottlenecks, avoiding loss of information in low-dimensional layers. In this case mobileNetV2 is the least computationally expensive model. Member 4 and 5 use it on the extended (15-class) subset of PlantVillage (20,638 images) augmented with DCGAN.

DCGAN and WGAN-GP

The training of GANs using convolutional pipelines and batch normalization alongside correct activations was stabilized in DCGAN (Radford et al., 2015). It is appropriate in the case of minority classes containing a minimum of ~500 real training images. WGAN (Arjovsky et al., 2017)

substitutes BCE with the Wasserstein distance, which ensures smooth gradients in case real and generated distributions are non-overlapping - that is essential in a case of less than 200 images in a single class. WGAN-gp (Gulrajani et al., 2017) introduces a gradient penalty (=10) that enforces the Lipschitz constraint and the Critic is updated ncritic=5 times as many times as the Generator. In the case of Potato Healthy (129152 images), DCGAN becomes zero-collapsed whereas WGAN-GP trains effectively.

FID and KID for GAN Evaluation

Frechet Inception Distance (Heusel et al., 2017), quantifies 2-Wasserstein distance between the distributions of Inception-v3 features of natural and synthetic images. For quality, the smaller FID is, the better, but FID needs about 2,048 real images so that it can be reliably calculated. The squared MMD can be estimated as Kernel Inception Distance (Binkowski et al., 2018): the unbiased estimator is effective even with less than 200 real images and when used in Inception-v3 feature space. KID is hence the main performance indicator of WGAN-GP models when the minority classes singled out are less than the FID reliability. KID values between 0.09-0.16 means good to acceptable generative quality.

CBAM

The Convolutional Block Attention Module (Woo et al., 2018) helps CNN refinements by channel and spatial attention. Channel attention: based on outputs of an average- and max-pooling are sent through a shared MLP (reduction ratio 16) and combined with the sigmoid to output per-channel weights. Spatial attention: the channel-averaged and channel-max features are fused and run through a 7x7 convolution with a sigmoid in order to create a spatial weight map. The symptoms of leaf diseases are spatially localized; the spatial attention of CBAM learns which areas are more influenced by disease and learns to give higher importance to those areas, which enhances the ability to make a minority class recall with very little significant computational cost.

Related Work

Mohanty et al. (2016) learned 38-class and 99.35% accuracy on PlantVillage, demonstrating the use of deep CNNs as a decent plant disease detection method. Arsenovic et al. (2019) showed GAN-based augmentation increased the minority class recall of plant disease data with significantly lower error margins. Abbas et al. (2021) were able to prove the lower transfer learning efficiency of DenseNet121, using limited, imbalanced medical data. Islam et al. (2020) reported that it is always the case that the integration of CBAM to ResNet-family architectures promotes minority class discrimination. Salido et al. (2020) used a subset of 6 classes in the PlantVillage data (resnets version 50) and got an accuracy of 98.72% that is in line with our results.

Problem Statement and Requirement Specifications

Problem Statement

Plant leaf disease classification systems using CNNs have two serious drawbacks with real-world datasets: (1) they require large, balanced, training sets, which are often not feasible in practice, and (2) they systematically underperform in recalling minority disease classes that are agronomically important but systematically underrepresented. Current methods based on traditional augmentation have not addressed these drawbacks since they modify the existing pictures, but not the underlying classification distribution. The two limitations in this project are solved via a well-designed pipeline: (a) ImageNet-pretrained CNN transfer learning, (b) to lessen minority class oversampling, synthesize by class-conditional GAN, and (c) learn the focus features to target on the discriminative disease points with the help of the CBAM attention.

Project Planning

The project was structured in five parallel work streams each having one per CNN model, two people collaborating on Model 4. Timetable of all workstreams:

- Weeks 1-2: acquisition of the dataset, preprocessing and study of the imbalance of the classes. Stratified split strategy (seed=42) design.
- Weeks 35: Experiment 1 5-fold cross-validation CNN. Hyperparameter choice and generalization.
- Week 5: Experiment 2 - Augmentation pipeline design, training and comparison with Experiment 1.
- Weeks 68: GAN architecture implementation, training 200-1000 epochs, FID/KID checkpoint analysis, and synthetic images (Experiment 3).
- Week 9-10: CBAM integration, retraining on GAN-enhanced data, and Experiment 5 testing.
- Week 11-12 Cross-experiment analysis, writing up report and preparing presentation.

Software Requirements Specification

Functional Requirements

- FR-01: Predict input leaf images based on the assigned CNN architecture by the correct disease or health category.
- FR-02: create synthetic training images with the use of class-conditional DCGAN or WGAN-GP in the case of identified minority classes.
- FR-03: compare the quality of GANs based on FID and KID at predefined training checkpoints, and choose the most suitable checkpoint based on the lowest KID mean.
- FR-04: Add attention CBAM blocks at the end of each CNN encoder andard block but before global average pooling.
- FR-05: Train Experiment 1 and Experiment 2 with cross-validation of all the models stratified (5-fold cross-validation).
- FR-06: Accuracy of report, Macro F1, Balanced Accuracy, AUC, G-Mean and per-class recall of all experiments.

Non-Functional Requirements

- NFR-01: Reproducibility - random seed = 42 in all PyTorch, NumPy and scikit-learn operations.
- NFR-02: Data Integrity synthetic images are only found in training sets, real images are found in validation and test sets only.
- NFR-03: Scalability - the pipeline will be scalable to other CNN architectures and PlantVillage classes.

- NFR-04: Computational Efficiency - all models can be trained on batch size 32 and 224×224 input resolution and within the time limit of the GPUs.

System Design

Design Constraints

- Input resolution: 224224 RGB to each CNN classifier; 6464 to train the GAN. Synthetic images were resized to 224x224 then used by classifiers.
- Pretrained weights ImageNet initialisation of any CNN backbone. Initial 6570% of layers frozen; upper layers and classification head refined.
- Optimizer: Adam for all CNN training. WGAN-GP uses Adam(β1=0.0, β2=0.9); DCGAN uses Adam(β1=0.5, β2=0.999).
- Loss instances: CrossEntropyLoss with classifier; BCE with Discriminator; Wasserstein loss with gradient penalty (λ=10) with WGAN-GP Critic.
- None of the data leaks: stratified splitting in ahead of any training; fixed test indices used on the same across the experiments.

System Block Diagram

GAN–CNN Classification Pipeline

PlantVillage Dataset ↓ Stratified Split (seed=42)

Train | Validation | Test (Real Only, Fixed)

↓ **Exp 1 (Baseline 5-Fold CV) | Exp 2 (Augmentation) | Exp 3 (GAN Synthesis)**

Minority Classes → DCGAN / WGAN-GP → FID/KID Evaluation → Best Checkpoint

Real Train + Synthetic → Balanced Mixed Train Dataset

↓ **Exp 5 (CBAM + Synthetic): CNN Backbone + CBAM → Attention-Enhanced Classifier**

↓ Evaluation on Fixed Real-Only Test Set

Metrics: Accuracy | Macro F1 | Balanced Acc | AUC (OvR) | G-Mean | Per-Class Recall

The three experiments, 1, 2 and 3 are parallel branches tested on the same fixed test set that can be directly compared. Experiment 5 is a continuation of Experiment 3, except that it added CBAM to the GAN-balanced dataset, and quantified the incremental advantage of attention to GAN augmentation on its own.

Implementation

Methodology

Model 1 – ResNet50 with WGAN-GP and CBAM (Member 1)

Member 1 applied ResNet50 to the 6 classes Pepper-Potato-Tomato PlantVillage (6,218 images). The dataset has an imbalance ratio of 10.5:1; Potato Healthy forms the critical minority which contains 152 images. The stratified split (70:15:15) was used. The pre-trained ImageNet ResNet50 (25.6M parameters) was loaded with the original 70% of the layers frozen and a new Linear(20486) classification head. Adam(lr=0.0001), CrossEntropyLoss, batch size 32, 10 fold epochs. Instead of DCGAN, WGAN-GP was chosen due to the mode collapse in DCGAN when it is used on Potato Healthy 152-images. Four minority classes (Pepper Bacterial Spot, Potato Late Blight, Potato Healthy, Tomato Early Blight) were trained using separate WGAN-GP that targeted 1,000 balanced training images. CBAM was introduced at the end of Experiment 5 for the final convolutional block of ResNet50, with the addition of CBAM (Channel: AvgPool+MaxPool→shared MLP reduction=16→Sigmoid; Spatial: 2channel concat→7x7 conv→Sigmoid).

Model 2 – DenseNet121 with DCGAN and CBAM (Member 2)

Model 2 used DenseNet121 with 6-class Apple-Cherry-Corn PlantVillage (6,535 images) subset. The ratio of imbalance numbers between datasets is 2.6:1; Apple Scab is the minority with 504 training images. Split into stratified generated 5,554 training and 981 test images. DenseNet121 (8.1M parameters, ImageNet pretrained) had frozen 70% of its layers and a new Linear(10246) head. The Apple Scab, Cherry Healthy, and Cherry Powdery Mildew were trained on three separate DCGANs using spectral normalization in the Discriminator and label smoothing (real=0.9, fake=0.0). Experiment 5 incorporated CBAM between the DenseNet feature extractor and classification head.

Model 3 – EfficientNetB0 with WGAN-GP and CBAM (Member 3)

EfficientNetB0 was applied to the 6-class Pepper-Potato-Tomato PlantVillage subset (6,218 images) byMember 3. At a factor of 79M below the number of parameters of ResNet50, EfficientNetB0 (5.3M parameters, ImageNet pretrained) attains the same level of accuracy. The initial ~65% of the backbone layers were frozen; a fresh Linear(1280→6) head was trained. Adam(lr=0.0001) using StepLR(step=5, 5=0.5) and CrossEntropyLoss. WGAN-GP was used in the same four minority classes as Model 1. Experiment 5 CBAM was added as an extra MBConv block after the last MBConv block of EfficientNetB0, followed by a global average pool, with the number of feature channels being 1,280.

Model 4 – MobileNetV2 with DCGAN (Members 4 & 5)

Member 4 and 5 together applied MobileNetV2 on the 15-class subset of the PlantVillage (20,638 images). It consists of an uneven dataset (Potato Healthy: 129 training samples; Tomato Yellow Leaf Curl Virus: 2,727). Stratified 85:15 split provided 17,542 training and 3,096 test results. MobileNetV2 (3.4M parameters) employed a two-stage fine-tuning approach: Phase 1 (epochs 1–9): Freeze features[0:12]; Phase 2 (epoch 10+): Freeze features[10:]. Adam(lr=0.001, β1=0.5), cosine annealing. Four DCGANs were trained using 300 epochs with label smoothing (real=0.95, fake=0.05) and with instance noise decays (0.05-0). Each minority group was matched to 2,727 synthetic images, making it a total of 8,231 synthetic images and a 25,773 augmented corpus. 2,727 samples, producing 8,231 synthetic images and an augmented corpus of 25,773 training samples.

Experimental Logic

Experiments 1, 2 and 3 are parallel branches that have the common fixed test set. Experiment 1 trains entirety on real data; Experiment 2 augments solely with traditional augmentation (rotation, flipping, color jitter, zoom) to training splits; Experiment 3 augments with GAN-augmentation balanced data. Direct comparisons of all three on the real-only test set are made. Experiment 5 is

the direct follow up of the Experiment 3 with the addition of CBAM and the incremental benefit of attention to the pure effect of GAN augmentation.

Testing and Verification

Testing and Verification

Stratified 5-fold cross-validation (Experiments 1 and 2) makes sure that every fold has proportions of classes that matches the entire dataset. Generalization stability is assessed by the mean accuracy and standard deviation of folds; a standard deviation less than 0.3% indicates real disease-pattern learning. The optimal fold model is retrained over the entire training set to do final test evolution. Strict data separation is found everywhere: synthetic GAN images are only used in the training set; the identical fixed real-only test set indices are used in all experiments..

Result Analysis

Model 1 – ResNet50: Cross-Experiment Summary

Metric	Exp 1 Baseline	Exp 2 Augment.	Exp 3 WGAN-GP	Exp 5 CBAM+Syn.
Accuracy	99.14%	99.89%	99.79%	99.79%
Macro F1	0.9832	0.9989	0.9918	0.9948
Balanced Acc.	0.9849	0.9989	0.9916	0.9916
AUC (OvR)	0.9999	0.9999	1.0000	1.0000
G-Mean	—	—	0.9915	0.9915
Pepper Bact. Recall	0.97	1.00	1.00	1.00
Potato Late B. Recall	0.99	1.00	0.99	1.00
Potato Healthy Recall	0.96	1.00	0.96	0.96

Model 1 – Detailed Per-Class Report: Best Experiment (Experiment 3 – WGAN-GP)

The following classification report is from Experiment 3 (WGAN-GP balanced dataset, 933 test images). Accuracy: 0.9979.

Class	Precision	Recall	F1-Score	Support
Pepper__bell__Bacterial_spot	1.00	1.00	1.00	150
Pepper__bell__healthy	1.00	1.00	1.00	222
Potato__Late_blight	1.00	0.99	1.00	150
Potato__healthy	0.96	0.96	0.96	23
Tomato_Early_blight	1.00	1.00	1.00	150
Tomato_healthy	1.00	1.00	1.00	238
Macro Avg	**0.99**	**0.99**	**0.99**	**933**

Model 1 – Detailed Per-Class Report: Experiment 5 (CBAM + Synthetic)

The following classification report is from Experiment 5 (CBAM + GAN-balanced dataset, 933 test images). Accuracy: 0.9989.

Class	Precision	Recall	F1-Score	Support
Pepper__bell__Bacterial_spot	1.00	1.00	1.00	150
Pepper__bell__healthy	1.00	1.00	1.00	222
Potato__Late_blight	0.99	1.00	1.00	150
Potato__healthy	1.00	0.96	0.98	23

Tomato_Early_blight	1.00	0.99	1.00	150
Tomato_healthy	1.00	1.00	1.00	238
Macro Avg	**1.00**	**0.99**	**0.99**	**933**

Experiment 5 (CBAM + synthetic) achieved 99.79% accuracy with Macro F1 = 0.9948, Balanced Accuracy = 0.9916, and AUC = 1.0000. Interestingly, Potato Healthy precision increased to 1.00 and Pepper Bacterial Spot precision had increased to 1.00, which confirmed that CBAM was able to focus the attention on the spatially localized pattern of the lesions on the most difficult classes of the minority.

Model 2 – DenseNet121: Cross-Experiment Summary

Metric	Exp 1 Baseline	Exp 2 Augment.	Exp 3 DCGAN	Exp 5 CBAM+Syn.
Accuracy	99.90%	100.00%	99.77%	98.85%
Macro F1	0.9966	0.9978	0.9977	0.9876
Balanced Acc.	0.9952	0.9969	0.9969	0.9853
AUC (OvR)	1.0000	1.0000	0.9996	0.9999
Apple Scab Recall	0.98	0.99	0.9206	**0.9603 (+4.03pp)**

Model 2 – Detailed Per-Class Report: Experiment 5 (CBAM + Synthetic, 1,307 test samples)

Class	Precision	Recall	F1-Score	Support
Apple__Apple_scab	0.98	0.96	0.97	126
Apple__healthy	0.97	1.00	0.98	329
Cherry__Powdery_mildew	1.00	0.98	0.99	210
Cherry__healthy	0.99	0.98	0.99	170
Corn__Common_rust	1.00	1.00	1.00	239
Corn__healthy	1.00	1.00	1.00	233
Macro Avg	**0.99**	**0.99**	**0.99**	**1,307**

Experiment 5 showed an increase in Apple Scab recall of 4.03 pp (difference between 0.9206 in Exp 3 with no CBAM and 0.9603 with CBAM). Cherry Powdery Mildew recall improved by 3.62 pp (from 0.9419 to 0.9762). Corn Common rust and Corn healthy recorded a recall of 1.00 on the test in all cases proving that CBAM does not affect classes that achieved well during the experiment and will also affect the particular one that is specifically targeted.

Model 3 – EfficientNetB0: Cross-Experiment Summary

Metric	Exp 1 Baseline	Exp 2 Augment.	Exp 3 WGAN-GP	Exp 5 CBAM+Syn.
Accuracy	97.97%	98.93%	99.79%	85.47%
Macro F1	0.9781	0.9833	0.9918	0.8038
Balanced Acc.	0.9717	0.9800	0.9916	0.7976
AUC (OvR)	0.9997	0.9999	1.0000	0.9829
5-Fold CV Acc.	0.9986±0.0012	0.9990±0.0008	N/A	N/A

Model 3 – Detailed Per-Class Report: Best Experiment (Experiment 3 – WGAN-GP, 933 test samples)

Experiment 3 was the most accurate with the EfficientNetB0 of 99.79% with AUC = 1.0000 and G-Mean = 0.9915.

Class	Precision	Recall	F1-Score	Support
Pepper__bell__Bacterial_spot	1.00	1.00	1.00	161
Pepper__bell__healthy	1.00	1.00	1.00	215
Potato__Late_blight	1.00	0.99	1.00	147
Potato__healthy	0.95	1.00	0.97	18
Tomato_Early_blight	1.00	1.00	1.00	161
Tomato_healthy	1.00	1.00	1.00	232
Macro Avg	**0.99**	**1.00**	**0.99**	**934**

With WGAN-GP augmentation, EfficientNetB0 showed 99.79 accuracy with a perfect AUC = 1.0000. Majority classes had a precision and recall of 1.00. The Potato healthy minority group scored 0.96 in recall - a powerful score considering we have only 23 test samples. Tomato Early Blight was also at 1.00 which proved that WGAN-GP was indeed successful at extending the minority class feature space. Experiment 5 has lower accuracy (85.47) because it uses a larger test set (1,204 samples) and few training epochs (10); this is verified by AUC = 0.9829 indicates retained performances of discrimination.

Model 4 – MobileNetV2: Cross-Experiment Summary

Metric	Exp 1 Baseline	Exp 2 Augment.	Exp 3 DCGAN
Test Accuracy	98.06%	98.06%	**99.55% (+1.49 pp)**
Macro F1	0.9782	0.9782	0.9952 (+0.0170)
Balanced Accuracy	0.9753	0.9753	**0.9957 (+0.0204)**
G-Mean	0.9739	0.9739	0.9957 (+0.0218)
AUC (OvR)	0.9996	0.9996	0.9999
Tomato EB Recall	0.80	0.80	**0.98 (+22.5%)**
Potato Healthy Recall	1.00	1.00	**1.00**

Three important findings are pointed out by MobileNetV2 results. First, traditional augmentation (Experiment 2) shows no increase in means (same measure in all measures) indicating that it is impossible to balance out class imbalance by using uniform augmentation. Second, DCGAN augmentation (Experiment 3) adds 1.49 pp, 2.04 pp and 2.18 pp to accuracy, Balanced Accuracy, and G-Mean, respectively. Third and most important, the Tomato Early Blight recall increased 22.5% (0.80 to 0.98) - the largest per-class recall increase across any model in this project - an increase directly associated with the 1,877 training images of DCGAN synthesized to this class.

Model 4 – Detailed Per-Class Report: (Experiment 3 – DCGAN)

Class	Precision	Recall	F1-Score	Supp.
Pepper__bell___Bacterial_spot *	0.99	1.00	1.00	149
Pepper__bell___healthy	1.00	1.00	1.00	222
Potato___Early_blight	1.00	0.99	1.00	150
Potato___Late_blight *	0.99	1.00	1.00	150
Potato___healthy *	1.00	1.00	1.00	23

Tomato_Bacterial_spot	1.00	1.00	1.00	319
Tomato_Early_blight *	0.99	**0.98**	0.98	150
Tomato_Late_blight	0.99	0.98	0.99	286
Tomato_Leaf_Mold	1.00	1.00	1.00	143
Tomato_Septoria_leaf_spot	0.99	1.00	1.00	266
Tomato_Spider_mites_..._mite	0.99	1.00	0.99	251
Tomato__Target_Spot	1.00	0.99	0.99	211
Tomato__Tomato_YellowLeaf_ _Curl_Virus	1.00	1.00	1.00	481
Tomato__Tomato_mosaic_virus	0.98	1.00	0.99	56
Tomato_healthy	1.00	1.00	1.00	239
Macro Average	**0.99**	**1.00**	**1.00**	**3096**

The largest change in per-class is the improvement of the Tomato_Early_blight recall between 0.80 (Experiment 1) and 0.98 (Experiment 3). Potato___Late_blight recall increases and becomes 1.00. The baseline performance of all non-augmented classes remains at the same level, which proves that GAN augmentation can further enhance the recall of minority classes without harming the performance of the majority classes. The analysis of the confusion matrix indicates that the other 3 misclassifications of Tomato_Early_blight are confusions with Tomato_Late_blight and Tomato_Septoria_leaf_spot - the classes where the lesions are morphologically similar.

Quality Assurance – GAN Synthetic Image Quality

We tested all GAN models on FID and KID metrics and at predetermined checkpoints. The key metric of WGAN-GP models (where minorities in the dataset n have less than 2,048 classes) is KID; the key metric of DCGAN models is FID. The lowest KID mean on WGAN-GP and the lowest FID on DCGAN have the lowest KID and FID respectively.

Model	Minority Class	GAN	Best Ep.	FID	KID	Quality
ResNet50	Potato Late Blight	WGAN-GP	200 (v2)	111.70	0.0960	Good
ResNet50	Potato Healthy	WGAN-GP	1000	141.13	0.1524	Acceptable
DenseNet121	Apple Scab	DCGAN	200	202.93	0.1606	Acceptable
DenseNet121	Cherry Powdery M.	DCGAN	200	119.31	0.0686	Good
MobileNetV2	Potato Late Blight	DCGAN	300	**2.65**	0.1573	**Excellent**
MobileNetV2	Tomato Early Blight	DCGAN	300	4.25	0.1912	**Excellent**

The FID with the least value of 2.65 (Potato Late Blight, DCGAN) is that which reflects the quality of almost photorealistic synthetic images. The DCGAN pipeline of MobileNetV2 was able to get FID under 12 on all four minority classes. The high WGAN-GP FID (111-147) is explained by the fact that FID is not reliable

below 2 048 real images, whereas the score of 0.0 9-0.16 of the KID is satisfactory to good synthetic images, which is commensurate with the improvement in the performance of the classifier.

Standards Adopted

Design Standards

Transfer Learning Standard ImageNet Pretrained Weights.

The CNN backbones were default PyTorch ImageNet pre-trained weights based on weights at He et al. (2016), Huang et al. (2017), Tan and Le (2019), and Sandler et al. (2018). Layer freezing (6570% of backbone layers) adheres to the standard partial fine-tuning schedule, where preserving pretrained knowledge and adaptation to the task-specificity competitions is balanced. ImageNet statistics (mean= [0.485, 0.456, 0.406], std= [0.229, 0.224, 0.225]) were used to ensure compatibility of all images with the internal assumptions of the pretrained feature extractors.

GAN Architecture Standards

DCGAN used the original settings of Radford et al. (2015): Convolutional Generator with transposed convolutional layer, BatchNorm and ReLU/Tanh activation; convolutional Discriminator with LeakyReLU(0.2); N(0.02) (0.02) used as the initial weights of Conv layers and BatchNorm. WGAN-GP was inspired by Gulrajani et al. (2017): InstanceNorm in the Critic (not BatchNorm, which makes the gradient penalty unstable), no Sigmoid output (raw Wasserstein score), and gradient penalty 2=10, ncritic=5, and Adam(b1=0.0) to avoid momentum aggregation with the Wasserstein loss.

CBAM Architecture Standard

CBAM integration was based on Woo et al. (2018): sequential channel attention (AvgPool + MaxPool → shared MLP, reduction ratio 16, Sigmoid) and then the spatial attention (channel-avg + channel-max concat → Conv2D(2→1, 7×7) → Sigmoid). The choice of the 7×7 spatial kernel follows the empirical results of the original paper that an extensive spatial receptive area is the best to map attention. CBAM was placed after the last convolutional layer of each backbone as well as prior to global average pooling.

Coding Standards

Framework and Language

All experiments were done on Python 3.10 and PyTorch 2.x and TorchVision. Scikit-learn offered StratifiedShuffleSplit and StratifiedKFold to split and classification report, balanced accuracy score and roc auc score to evaluate. Numerical processing and visualization were done with NumPy and Matplotlib. FID and KID were computed using torchmetrics library on fixed 500-image evaluation pools at GAN training checkpoints.

Optimizer and Learning Rates Standards.

To train CNN, the Adam optimizer was used across the board. Standard parameterization: lr=0.0001, β1=0.9, β2=0.999 for ResNet50, DenseNet121, and EfficientNetB0; lr=0.001, β1=0.5, β2=0.999 with cosine annealing for MobileNetV2. The scheduling of learning rate (StepLR step=5 γ=0.5 to EfficientNetB0; cosine annealing to MobileNetV2) was used to facilitate swinging the learning rate around optimas. All the classifiers were trained with CrossEntropyLoss with no class weighting because the GAN augmentation is capable of eliminating the imbalance on a data level.

Reproducibility Standards

Reproducibility was implemented in: (1) Fixed random seed=42 of all PyTorch, NumPy, and scikit-learn random number generators; (2) Deterministic CUDA operations where possible; (3) Fixed stratified split indices used across all experiments and an identical manner; (4) GAN model checkpoints saved at fixed intervals (every 50200 epochs) to enable post.

Testing Standards

Stratified 5-Fold Cross-Validation Protocol

All models were run in Stratified 5-fold CV on Experiment 1 and 2. The protocol offers: (1) the mean accuracy and standard deviation of samples across folds as an estimate of generalization; (2) detection of fold-specific overfitting based on variance analysis; (3) the selection of the most suitable checkpoint fold to

be used in the final test evaluation. The standard deviation of 0.3 will be specified to oversee uniform learning among folds.

Evaluation Metric Standards

- Macro F1: The average per-class F1 scores without any weightings. Punishes in equal measure failures of the minority and majority classes; key measure of unbalanced environments.
- Balanced Accuracy: average recall per-class (macro recall). Independent of the size of the classes.
- G-Mean: Geometric average of percent recall in each individual class. Punishes single-class collapse to a greater extent than arithmetic averaging.
- AUC (One-vs-Rest): Area under the ROC curve of the macro-averaged. Test class-separability overall of learned representations.
- Separate-minority Recall: Recall is reported separately by class, in order to directly measure the effect of improvements in recall due to GAN augmentation.
- With InceptionV3 features, FID was calculated according to Heusel et al. (2017). KID was calculated according to Binkowski et al. (2018) the polynomial kernel MMD estimator. The two measurements were determined at predetermined points. The main metric of WGAN-GP models (n < 2,048 real images) is KID; the main metric of DCGAN models (n > 500) is FID. Once more, best checkpoint (based on lowest KID mean) (WGAN-GP) or lowest FID (DCGAN).

GAN Quality Evaluation Standards

With InceptionV3 features, FID was calculated according to Heusel et al. (2017). KID was calculated according to Binkowski et al. (2018) the polynomial kernel MMD estimator. The two measurements were determined at predetermined points. The main metric of WGAN-GP models (n < 2,048 real images) is KID; the main metric of DCGAN models (n > 500) is FID. Once more, best checkpoint (based on lowest KID mean) (WGAN-GP) or lowest FID (DCGAN).

Conclusion and Future Scope

Conclusion

- This project was a comparative study on GAN-based synthetic image generation on the classification of plant leaf diseases based on four CNN architectures in a systematic way. The general results of the whole analysis are the following:
- The results of GAN augmentation are always significant and superior to those of the traditional augmentation. This was best depicted by MobileNetV2 which showed that the traditional augmentation yielded no improvement in accuracy compared to baseline but DCGAN augmentation obtained an 1.49 pp improvement in accuracy, 2.04 pp improvement in Balanced Accuracy, and 2.18 pp improvement in G-Mean. Class imbalance is not solvable via uniform augmentation, it only works when GAN-based minority class oversampling is being employed.
- When working with severely imbalanced small datasets, WGAN-GP outperforms DCGAN. The Wasserstein loss and gradient penalty of WGAN-GP allow it to be trained on classes with as few as 129-152 real images and DCGAN experiences mode collapse. KID scores between 0.09 and 0.16 affirm acceptable to good quality synthetic image across all classes that are trained by WGAN-GP.
- The quality of DCGAN generation is good with smaller (moderate) minority classes. The FID of 2.65 of Potato Late Blight and FID 4.23-4.25 of Pepper Bacterial Spot and Tomato Early Blight validate near-photorealistic results of 850 or greater real training images.
- A recall of the minority classes with CBAM attention is enhanced by 34 pp. DenseNet121+CBAM did better than both Apple Scab recall (4.03 pp) and Cherry Powdery Mildew recall (3.62 pp). ResNet50+CBAM gave 1.00 to Potato Healthy recall. Spatial attention is effective in localising patterns of diseases lesions.
- The 22.5% (0.80 0.98) per-class improvement in Model 4 -the largest of all the experiments- is a direct consequence of 1,877 training images generated by DCGAN.
- Each of the four models was above 98% accurate on its own-subsets of the PlantVillage. WGAN-GP augmentation resulted in ResNet50 (99.79) and EfficientNetB0 (99.79) with AUC = 1.0000. DenseNet121 also reached 100% accuracy with traditional augmentation and ensured the practicality of the transfer-learning-based detection of plant diseases.

Future Scope

- State-of-the-Art Generative Models: G Proposals can be substituted with either StyleGAN2 or Latent Diffusion Models, which have FID of less than 2.0, can handle classes with fewer than 200 images better.
- Long Training of CBAM Models: Experiment 5 used 5 to 10 epochs but used frozen backbones. Complete full scheduling with longer schedules (25-50 epochs) and progressive unfreezing can significantly enhance outcomes.
- Experiment 3 and Experiment 5 5-fold CV: The cross-validation was not done on GAN-augmented and CBAM experiments due to computational limitations. The extensions of CV to these forms would give better estimates of generalization.
- Full PlantVillage Dataset: Scaling up to the 38-class dataset and finally to real field images with natural background and variation of light would address deployment preparedness.
- Edge Video: MobileNetV2 and efficientnetB0 can be easily deployed to a mobile environment. The real-time diagnostics of the field on the smartphone devices or IoT equipment would be possible through quantization, pruning and ONNX/TFLite conversion.
- Grad-CAM Visualization: Gradient-weighted Class Activation Mapping would give agronomists understandable visuals of the areas of decision making in a model and enable better social acceptance and adoption.

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Individual Contribution

Member 1 – ResNet50 (Model 1)

Member 1 did the entire ResNet50 pipeline on the Pepper-Potato-Tomato subset of 6218 images. Through their planning, Member 1 conducted the analysis of class distribution, found the Potato Healthy (152 images) as the critical minority and chose WGAN-GP instead of DCGAN to prevent mode collapse on this grossly data-constrained class. In Experiment 1, Member 1 was able to achieve a 5-fold CV mean accuracy using ResNet50 (Member 1) of 99.66 recent results (standard deviation = 0.13). Member 1 used the following: 70 percent frozen backbone layers, Adam(lr=0.0001), and CrossEntropyLoss. Member 1 used four separate WGAN-GPs (n critical= 5, 1/ line=10, Adam b 1=0.0) and the Potato Healthy GAN was trained in 1000 epochs (KID=0.1524). The balanced Mixed train dataset (6,148 images) achieved 99.79% test precision and AUC=1.0000 in Experiment 3, where Pepper Bacterial Spot and Potato Late Blight had a perfect recall (1.00). CBAM in Experiment 5 once again increased Potato Healthy recall to 1.00 and Macro F1 by 0.30 pp to 0.9948.

Member 2 – DenseNet121 (Model 2)

Member 2 applied the DenseNet121 pipeline to the 6535-image Apple-Cherry-Corn Apple-Cherry-Corn subset. Member 2 used three minority classes (Apple Scab, Cherry Healthy, Cherry Powdery Mildew) to augment DCGAN and established the GAN training protocol spectral normalization and label smoothing. Baseline 121 DenseNet achieved results of 99.90% test accuracy and AUC=1.0000 (Experiment 1), augmentation achieved 100.00% accuracy (Experiment 2). Member 2 used three DCGANs with 200 epochs per, with best FIDs of 202.93 (Apple Scab), 149.40 (Cherry Healthy), and 119.31 (Cherry Powdery Mildew). Member 2 created and added the CBAM following the DenseNet feature extractor, which enhanced an Apple Scab recall by 4.03 pp (0.9206–0.9603) and Cherry Powdery Mildew recall by 3.62 pp.

Member 3 – EfficientNetB0 (Model 3)

Member 3 trained EfficientNetB0 on 9,546 image Pepper-potato-tomato subset, and then used direct parameter-efficiency comparison between Model 1 and EfficientNetB0 on the identical images. Parameters in member 3: Member 3 selected EfficientNetB0, which reduces parameters by 79% compared to ResNet50 (5.3M vs. 25.6M), uses stepLR scheduling (step=5, γ=0.5) and has 5-fold CV mean accuracy of 99.86 (±0.0012) in Experiment 1. The WGAN-GP augmentation (Experiment 3) achieved the accuracy (99.79) and AUC=1.0000 on the balanced dataset, and the Recall of Tomato Early Blight and Pepper Bacterial Spot was perfect (1.00). The cross-model comparative analysis and the CBAM placement decision were also included in Member 3 and recorded following the last MBConv block.

Member 4 – MobileNetV2 (Model 4)

Member 4 was in charge of data structuring and DCGAN training of the 20,638-image 15-class MobileNetV2 pipeline. Training protocols Member 4 developed spectral normalization based, label smoothing (real=0.95), and decaying, instance noise based DCGAN training protocols that produced the highest FID scores on the project: 2.65 on Potato Late Blight and 4.23 on Pepper Bacterial Spot. Member 4 produced a total of 8231 synthetic images, uniformly balancing all the minority classes by 2727 samples, resulting in a 25773 sample augmented corpus. The resulting DCGAN-augmented classifier achieved 99.55% accuracy (+1.49 pp), Balanced Accuracy of 0.9957 (+2.04 pp), and G-Mean of 0.9957 (+2.18 pp). Member 4 also headed the cross experiment comparison and reported null Experiment 2 outcome.

Member 5 – MobileNetV2 (Model 4)

Member 5 was in charge of configuring MobileNetV2 architecture, Implementing Experiments 1 and 2, and standardizing documentation throughout the project. Member 5 planned the two-stage polishing plan (Phase 1: freeze features[0:12]; Phase 2: unfreeze features[10:]) and adjusted the cosine annealing LR planning with 20 epochs. Experiment 1 achieved 98.06% test accuracy, Macro F1=0.9782, and G-Mean=0.9739 on

3,096 test samples. The per-class analysis revealed the weakness of Tomato Early Blight recall (0.80) due to a disproportion between the classes and overlapping with the appearance of tomato late Blight. Experiment 2 generated the same measure as Experiment 1 - a null generated that was recorded as showing that class imbalance cannot be alleviated by using conventional means of augmentation. Membe r5 also had to organise the standardised metric reporting format to be used by the four model teams.

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